MIL-R-36554
7 February 1968

MILITARY SPECIFICATION

REFRIGERATOR, MECHANICAL; BIOLOGICALS, AND WHOLE BLOOD, PORTABLE, FIELD

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of lightweight, portable, mechanical refrigerator of 2.85 cubic feet capacity for field use.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-C-310	Chlorofluoro Hydrocarbons (of the Methane and Ethane Series).
QQ-A-200/3	Aluminum Alloy Bar, Rod, Shapes, and Tube Extruded, 2024.
QQ-A-225/6	Aluminum Alloy Bar, Rod, and Wire; Rolled, Drawn, or Cold Finished, 2024.
QQ-A-250/2	Aluminum Alloy, 3003, Plate and Sheet.
QQ-A-250/8	Aluminum Alloy, 5052, Plate and Sheet.
QQ-A-250/11	Aluminum Alloy, 6061, Plate and Sheet.
QQ-S-561	Solder, Silver.
TT-E-529	Enamel, Alkyd, Semi-Gloss.
UU-P-271	Paper, Wrapping, Waterproofed Kraft.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.

Military

MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.

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STANDARDS

Federal

FED-STD-406	Plastics, Methods of Testing.
FED-STD-595	Colors.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.

DRAWINGS

Defense Personnel Support Center (DPSC)

10191	Refrigerator, Mechanical, Biologicals, Whole Blood.
24006	Rack, Blood Bottle and Bag.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following documents for a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society Testing Materials

ASTM D638	Method of Test for Tensile Properties of Plastics.
ASTM D1706	Method of Test for Indentation Hardness of Plastics by means of a Durometer.

(Application for copies should be addressed to American Society Testings Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

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National Fire Protection Association

NFPA 70 Standard for Electric Wiring and Apparatus.

(Application for copies should be addressed to National Fire Protection Association, 60 Batterymarch Street, Boston, Mass. 02110.)

Underwriters' Laboratories, Inc. (UL)

UL 303 Condensing Units, Refrigeration.

(Application for copies should be addressed to Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, Ill. 60611; 2550 Dundee Road, Box 247, Northbrook, Ill. 60062; 1285 Walt Whitman Road, Melville, L.I., N.Y. 11746; 1655 Scott Boulevard, Santa Clara, Calif. 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit to the contracting agency proof, prior to delivery, that the electrical components supplied under this specification conforms to the applicable requirements of Underwriters' Laboratories, Inc. Standard UL 303 and the National Fire Protection Association Standard NFPA 70. Such proof shall consist of a laboratory report or listing by the Underwriters' Laboratories, Inc. The label or listing of the Underwriters' Laboratories shall be acceptable as evidence that the equipment conforms to the above. A copy of each laboratory report shall be furnished to the contracting officer.

3.1.2 In lieu of the label or listing by the Underwriters' Laboratories, Inc., the contractor may submit a laboratory report by a nationally recognized testing agency adequately equipped and competent to perform such services, as shall be evidenced by the attachment of its seal, stamp or label to the article, or the issuance of a certificate. The contractor shall submit the laboratory report which shall include test results and methods used to determine compliance, to the contracting officer for approval. The Government reserves the right to require a facilities report showing the qualifications of the testing laboratory and to conduct a facilities survey to determine its acceptability.

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3.1.3 Compliance with the above requirements regarding fire and casualty hazards, does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.2 First article. Unless otherwise specified, prior to the start of production of the contract quantity, one sample of the refrigerator shall be assembled, completely finished, and tested by the contractor. The sample shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first article, acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.3 Material.

3.3.1 Aluminum. Aluminum alloys 5052 and 3003 shall conform to QQ-A-250/8 and QQ-A-250/2, respectively. Aluminum alloy 2024 shall conform to QQ-A-200/3 for extruded shapes and QQ-A-225/6 for rolled or drawn shapes. Aluminum alloy 6061 shall conform to QQ-A-250/11.

3.3.2 Plastic. The top cover for the shell, the shield for the top cover, and the spacers shall be odorless black thermosetting plastic.

3.3.3 Gasket. Gasket shall be a magnetic vinyl.(see 6.6).

3.3.4 Electrical components. Electrical components such as relay, relay receptacle plug, cord, plugs and receptacle, switch, heating element, and thermostatic switches shall conform to applicable portions of NFPA 70. The electrical power cord shall be 2 conductor, No. 16 wire, type SJO. Connecting wire shall be No. 18 stranded vinyl coated wire having a nylor jacket. Electrical components shall pass the dielectric test specified in 4.4.2.

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3.3.5 Refrigerant. Refrigerant shall be dichlorodifluoromethane conforming to type 12 of BB-C-310.

3.3.6 Strainer. Strainer wire shall consist of a copper screen wire encased in a copper housing tube.

3.3.7 Dryer. Dryer shall be the hermetic solder-connection type, with mesh screen at each end and silica gel fill. There shall be a 1/4 inch outside diameter connection at each end.

3.3.8 Silver solder. Silver solder shall contain at least 40 percent silver and conform to QQ-S-561.

3.4 Construction. The refrigeration unit shall consist of a refrigerator cabinet, a mechanical unit for cooling the storage compartment of the cabinet, and an electrical heater. The heater shall be of the hermetically sealed or moisture-proof type, and shall be bonded to the undersurface of the storage compartment. The unit shall be constructed in accordance with DPSC Drawing No. 10191. The cabinet shall have flush countersunk hardware, controls and thermometer. The refrigerator shall be capable of being easily secured and detached for transportation.

3.4.1 Refrigerator cabinet. The refrigerator cabinet shall consist of an insulated enclosure, a noninsulated machine compartment, and an insulated lid. The cabinet shall be constructed from an aluminum alloy fabricated shell to which shall be butt welded an aluminum alloy bottom. An aluminum alloy partition shall separate the insulated section from the machine compartment. The storage compartment shall be formed from aluminum alloy sheeting to which shall be welded an aluminum alloy bottom. Partition and shells shall be 0.064 inch thick, 5052 aluminum alloy. Hardwood supports shall maintain the storage compartment in position. The space between the storage compartment and the outer shell and partition shall be filled with four inch thick insulation. The cabinet shall be provided with two flush-mounted carrying handles, access panels for servicing the machine section and heater, and a front access through a hinged door for servicing the electrical components. The top cover for the shell shall be a 3/32 inch thick black thermosetting plastic sheet. The rim of the freezing compartment shall be sealed with a gasket (see 3.3.3). An aluminum

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alloy lid shall be hinged to the cabinet shell. An aluminum alloy arm shall hold the lid in the open position. The inside of the lid shall be completely filled with insulating material. The lid latch shall be constructed in accordance with drawing 10191. At no point shall the screws or bolts used in the construction or attachment of parts be so arranged that they will form a path of heat or moisture from the exterior to the interior. The partition separating the motor compartment from the storage compartment shall be sealed with a heavy application of a mositureproof and vaporproof compound. Plastic panels shall be similarly sealed to the cabinet body and cabinet lid.

3.4.2 Mechanical unit. The mechanical unit shall consist of a hermetically sealed compressor, a condenser forming part of the outer shell of the cabinet, an evaporator forming part of the outer shell of the storage compartment, and a temperature control providing selection of both a cabinet temperature of 20° F. and one of 37° F.

3.4.2.1 Compressor. The compressor shall be of the reciprocating type, hermetically sealed, designed for operation on 110 volts, 60 cycles, A.C., and shall have a rating of not less then 1/9 horsepower (h.p.). The compressor unit shall be furnished with an overload cutout. The compressor shall be mounted on angle supports.

3.4.2.2 Condenser. The condenser shall consist of 1/4-inch-diameter, continuous length, all copper tubing, attached to the inner wall of the outer shell of the cabinet. The tubing shall be silver soldered to the compressor outlet and the capillary tubing. The tubing may be attached to the shell by continuous straps spot welded in position, or may be furnace fused. The tubing shall be without a splice.

3.4.2.3 Evaporator. The evaporator shall consist of 1/4-inch-diameter, continuous length, all copper tubing, attached to the outer wall of the storage compartment. The tubing shall be silver soldered to the compressor inlet and capillary tubing. The tubing may be attached to the wall by continuous straps, spot welded in position, or may be furnace fused. Exterior parts of the evaporator tubing shall be covered with insulation tape to prevent condensation. The tubing shall be without a splice.

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3.4.2.4 Capillary tubing. The capillary tubing shall be 138 inches long with a bore of 0.031 inch. Variations from this size and length may be made, provided proper balance is maintained between the condenser and evaporator to meet the performance specified in 3.5. A length of 48 inches of this capillary tubing shall be soldered to the copper suction line of the compressor to act as a heat exchanger.

3.4.2.5 Dryer. A dryer, to act as a dehydrator and liquid receiver, shall be located in the compressor suction line as shown in the development of the evaporation coils on Drawing 10191. Silver solder shall be used on dryer solder connections.

3.4.2.6 Strainer. A strainer shall be located between the outlet of the condenser and the capillary tubing.

3.4.3 Controls. The temperature controls shall consist fo two hermetically sealed thermostats which control the compressor through a hermetically sealed relay and two hermetically sealed thermostats which control the operation of the heating element. The controls shall be so set that the heating system and the refrigeration systems shall not operate at the same time. The controls shall be located in a thermoswitch box on the machine compartment end of the cooling compartment. The refrigeration thermostat switches shall be preset to provide a temperature of 37° F. ± 1-1/2° F. and 20° F. ± 1-1/2° F. in the center of the cooling compartment and shall be selected by operating a two-position switch. The thermostats for controlling the heater shall be adjusted so that the heater shall be activated within 3 degrees of the lowest load temperature for each setting. The heating element shall be located below the storage compartment. The wiring diagram for the system shall be in accordance with DPSC Drawing No. 10191.

3.4.4 Relay. Each refrigerator shall be furnished with one spare hermetically sealed relay.

3.4.5 Dial thermometer. A dial thermometer shall be mounted on the front of the cabinet with the thermometer bulb extending into the thermoswitch box. The thermometer assembly shall be in accordance with Drawing No. 10191 and shall be replaceable from outside the cabinet.

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3.4.6 Refrigerant. Refrigerant shall be as specified in 3.3.5. The amount of charge shall be such that the proper balance shall be maintained between the condenser and evaporator up to an ambient temperature as high as 120° F.

3.4.7 Racks. Each refrigerator shall be furnished with three wire racks. Each rack shall be suitable for holding 12 blood bottles or 20 plastic blood bags (see 6.1) filled with 450 milliliters of blood. Racks shall be in strict accordance with Drawing 24006. Particular accuracy shall be maintained in overall dimensions of the rack and the storage compartment of the refrigerator to insure proper fit of the rack within the compartment. The racks shall stack. All wire ends and surfaces shall be rounded off, free of burs. All welds shall be ground smooth. The racks shall be coated with white cellulose acetate butyrate 0.012 to 0.015 inch thick (see 6.4). The plastic shall contain no plasticizers capable of extraction or migration which might deleteriously affect the blood bag. The minimum physical properties of the plastic shall be as listed below when tested as specified in 4.4.3.

Hardness Rockwell-R Scale	75
Tensile strength (psi)	4050
Ultimate elongation (percent)	66

In wear resistance, the coating shall not lose more than 60 milligrams per 1000 cycles when tested as specified in 4.4.3.3. The plastic coating applied to the metal substrate shall bond so that when the plastic layer is incised to the base metal, in two parallel lines 1/8 inch apart and 1 inch long, the plastic material within these incisions cannot be removed in one piece by prying and pulling with a knife blade.

3.5 Performance. The unit shall be capable of operating over an ambient temperature range of -60° F. to +120° F.

3.5.1 Refrigeration system. When tested as specified in 4.4.1.1, the refrigeration system shall have a capacity to cool the storage compartment, without load, from a soaked-out condition of 120° F. to a temperature of 20° F. at the center of the storage compartment in not more than 60 minutes. After the unit has reached cycling condition under 120° F. ambient, the running time at the 20° F. setting shall not exceed 60 percent of the cycle time. The maximum power requirement to accomplish the pulldown to 20° F. shall not exceed 220 watts. The refrigerator shall maintain temperatures of 20° ± 1.5° F. and 37° ± 1.5° F. through 2 cycles at a 120° ambient temperature.

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3.5.2 Heating system. When tested as specified in 4.4.1.2, the fully loaded refrigerator shall maintain 20° ± 1.5° F. and 37° ± 1.5° F. at the center of the load through 2 cycles at an ambient temperature of -60° ± 2° F. The heater shall be activated within 3 degrees of the lowest load temperature at each setting.

3.5.3 Control system. When tested as specified in 4.4.1.3, the refrigerator without load and at an ambient temperature of 60° to 90° F. shall maintain 20° ± 1.5° F. and 37° ± 1.5° F. throughout 5, eight hour cycles.

3.6 Finish. All exterior surfaces, including the hinges, the handle and the compressor compartment, shall be thoroughly cleaned and chemically degreased. They then shall be given one coat of zinc-chromate primer conforming to MIL-P-8585, followed by two coats of baked-on olive drab synthetic semi-gloss enamel conforming to TT-E-529. Each coat shall be sprayed on and baked in high-temperature ovens in accordance with the paint manufacturer's recommendations. The resultant finish shall be hard, smooth, and highly resistant to abrasion, chipping, and cracking. The storage compartment of the cabinet shall be unpainted. Color shall be olive drab, No. X24087 (see 6.2) of Federal Standard No. 595.

3.7 Identification marking. On the inside of the lid, the plastic sheet shall be engraved with the name or registered trademark of the manufacturer, serial number, Federal stock number, contract or order number and the letters U.S. As an alternate, attachment of a metal identification plate in accordance with MIL-STD-130 shall be acceptable. A metal identification plate containing the following information shall also be attached to the front of the refrigerator:

CAUTION: TO PREVENT OIL FLOW INTO THE DRYER, KEEP UPRIGHT.

NOTE: SECURE LID LATCH DURING TRANSPORTATION.

All markings shall be neat, legible and permanent. Marking secured by adhesives shall not be used.

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3.8 Service data. The contractor shall furnish, with each unit, two copies of a service data manual. In addition the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. This service data shall contain complete instructions for installation, operation, maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. Directions for replacing the dial thermometer from outside the refrigerator shall be included. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the date of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

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3.9 Workmanship. Workmanship shall be first class in every respect. The refrigerator shall be free from imperfections which detract from its appearance or may impair its serviceability. Refrigerator shall be free of sharp edges. Control box cover shall fit flush. Racks shall fit the compartment freely but without play.

3.9.1 Welding. All welding shall be done with appropriate equipment properly regulated and operated by thoroughly trained and experienced welders, highly skilled in the operations of aluminum welding. Seam and fillet welds shall be continuous, even, and smooth. Spot welds shall be accomplished so as to insure sound welds of sufficient strength to tear the spot out of the parent metal when pulled apart. All exposed seam and fillet welds shall be ground smooth to eliminate traces of weld marks, and shall be polished to blend with adjacent surfaces. All exposed spot welds shall be ground smooth, or filled and sanded smooth so that they shall not be visible after the enamel finish is applied.

3.9.2 Wiring. Wire terminals shall be staked on and insulated. Wiring shall be neatly laced or harnessed. All thermostat connections shall be inside the control box.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right ot perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. The unit of product for inspection purposes shall be one refrigerator. For inspection of the rack only, the unit of product shall be one rack.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table I. Acceptance number shall be zero for all tests.

Table I.

Component	Requirement	Test method	Inspection level
Refrigerator	Refrigeration system (3.5.1)	4.4.1.1	S-4
Refrigerator	Heating system (3.5.2)	4.4.1.2	2 units
Refrigerator	Control system (3.5.3)	4.4.1.3	S-4
Refrigerator	Dielectric strength (3.3.4)	4.4.2	100% inspection

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Table I. (cont)

Component	Requirement	Test method	Inspection level
Plastic on rack	Hardness (3.4.7)	4.4.3.1	*
Plastic on rack	Tensile strength (3.4.7)	4.4.3.2	*
Plastic on rack	Ultimate elongation (3.4.7)	4.4.3.2	*
Plastic on rack	Wear resistance (3.4.7)	4.4.3.4	*

* Sufficient lengths of plastic, 0.012 to 0.015 inch thick shall be used for tests.

4.3 Examination. The refrigerator shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Refrigerator not complete with all required parts, subassemblies, and materials.

102 Components and parts not free of cracks, tears, ruptures, and deformed surfaces.

103 Components, mating parts, and subassemblies not properly aligned and fitted.

104 Seam or fillet welds not sound, smooth, and free of flux, spatter, and discoloration.

105 Spot welds not sound and free of flash and discoloration.

106 Exposed or accessible edges, corners, and surfaces not free of burs, sharpness, and sharp projections.

107 Lid fails to seat properly on body and gasket to properly seal off storage compartment.

108 Lid fastner fails to latch and unlatch properly.

109 Cover hinge fails to operate smoothly and easily when cover is opened and closed.

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Major (cont)

110 Aluminum arm fails to hold cover securely in the open position.
111 Switchbox door fails to function properly.
112 Components in the electrical system not in accordance with requirements.
113 Electrical systems in units not consistent in layout and neatly wired.
114 Insulation of wire connectors not clear of, or protected from, framework edges to avoid occurrence of shock hazard or shorts due to vibration.
115 Electrical wiring not free of cracks, breaks, and cuts in the insulation.
116 Electrical system not free of taut leads, loose or improperly fastened terminal connections, and poor solder joints.
117 Racks fail to stack properly.
118 Racks fail to fit freely into storage compartment.
119 Rack not free of burs, sharp protruding edges, and cracked, discolored, and unsound spot welds.
120 Rack not properly and not uniformly clad.
121 Thickness of metal and plastic sheets for component parts and cross-sectional dimensions of tubing not within required limits.
122 Plastic cover for shell, lid and spacers not sealed to mating surfaces.

Minor

201 Overall and compartment dimensions not within specified limits.
202 Spacing between spot welds not at, or less than, distances set forth on the drawing.
203 Exterior parts of evaporator tubing not adequately or completely insulated.
204 Fasteners not securely tightened.
205 Enameled surfaces not free of blisters and chipped-out, cracked, nonadherent, bare, scuffed, and scratched areas.
206 Dial thermometer not properly calibrated.
207 Marking not present, properly located, legible, correct, or permanently affixed.
208 Service data not present in duplicate.

* Examination shall not be restricted to the classified possible defects listed above.

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4.3.2 Dimensional examination. The refrigerator shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted, to determine compliance with specification requirements.

4.4.1 Performance. The refrigerator shall be operated on a 115 volt, 60 cycle alternating current source. A potentiometer with thermocouples, and a wattmeter shall be used respectively to measure temperature and power consumed.

4.4.1.1 Refrigeration system. The refrigerator shall be soaked out with cover open, without load, at an ambient temperature of 120° F. ± 2° F. for a minimum of 12 hours. At the end of this time and while still at the 120° F. ambient, the cover shall be closed and the temperature switch set to 20° F. The thermocouple shall be located at the center of the storage compartment. The pull down time to reach 20° F., the power consumed by the unit for pull down, and running time while cycling at 20° F. shall be recorded. While the refrigerator is cycling at the designated 20° F. setting, the temperature in the storage compartment shall be recorded. The switch shall then be changed to the 37° F. setting and the temperature recorded while the refrigerator is cycling at the 37° F. setting. Temperature measurements shall be made through two complete cycles.

4.4.1.2 Heating system. The refrigerator racks shall be loaded with blood bags or bottles filled with SAE No. 10 lubricating oil. The refrigerator shall be operated at room temperature at the 20° F. setting until the temperature of the load is at equilibrium. The refrigerator shall then be operated in an ambient temperature of minus 60° F. The load temperatures shall be recorded until the heating system is cycling at the designated temperature setting and the load temperature is at equilibrium. The switch shall then be changed to the 37° F. setting and the refrigerator operated in an ambient temperature of minus 60° F. The load temperatures shall be recorded as specified for the 20° F. setting. The temperature of the oil in the bag or bottle near the center of the middle rack shall be measured by suitable means, such as thermocouple (see 6.3). Temperature measurements shall be made for a minimum of 2 complete cycles.

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4.4.1.3 Control system. The refrigerator shall be operated without load at room temperature (between 60° F. and 90° F.) for a total of 16 hours in two stages of 8 hours each, with a shutdown period of sufficient duration to assure that the units have been soaked out to ambient temperature. One stage shall be run with the temperature control set at 20° F. and the other with the temperature control set at 37° F. The accuracy of the controls shall be measured at both the 20° F. and 37° F. settings for a minimum of 5 cycles, using a thermocouple located at the center of the cooling compartment.

4.4.2 Dielectric test. The dielectric test shall be performed on the refrigerator with all current-carrying parts in both the refrigeration and the heating circuits, at both the 20° F. and the 37° F. settings. A test potential of 900 volts shall be developed between the current-carrying parts and the non-current-carrying parts of the refrigerator and maintained for 1 minute.

4.4.3 Plastic properties tests.

4.4.3.1 Hardness. Indentation hardness of the plastic shall be determined in accordance with ASTM Method D1706.

4.4.3.2 Tensile strength and ultimate elongation. Tensile strength and ultimate elongation shall be determined in accordance with ASTM Method D638.

4.4.3.3 Wear resistance. Wear resistance shall be determined by Method 1091 of Federal Standard No. 406 utilizing a Taber abrader with CS-17 calibrase wheels subjected to a 1000 gram load.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Level A.

5.1.1 Unit of issue. One refrigerator, as specified, constitutes one unit of issue.

5.1.2 Unit packaging. All exterior finished surfaces shall be protected by wrapping with kraft paper and cushioning with excelsior pads or flexible corrugated fiberboard or other suitable materials. Wrapping and cushioning material shall be secured by typing with cotton cord or tape, or by sealing with kraft gummed paper tape. Work shall be accomplished in such a manner as to insure against marring, scratching, or damage to the unit. Compressor unit shall be immobilized.

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5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Level A. Each unit shall be overpacked in an exterior container designed for type 3 load, and constructed in accordance with PPP-B-601, table II, using type III, class 1 plywood, or PPP-B-621, table IV. The unit shall be adequately secured within the exterior container by means of bolts, wood braces, or other interior fittings, to insure against shifting or breakage. Each unit shall be protected by a shroud fabricated of waterproof barrier material conforming to UU-P-271, class L-2(a) and L-2(b). The exterior container design shall include three skids of not less than 1-1/2 inches in thickness applied crosswise of the base, so positioned as to facilitate handling with a forklift truck and to maintain unit in an upright position. Closure and strapping shall comply with appendix of applicable specification. As an alternate, each unit shall be overpacked in a box constructed in accordance with PPP-B-640, class 2, style E. Three (3) wooden skids shall be nailed to the bottom of the box. Closure shall be in accordance with 30.3.5, except that use of steel strapping is not authorized. Sealing shall be in accordance with 30.2 by use of tape as specified in 30.1.6.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Exterior containers. Exterior containers shall be marked in accordance with MIL-STD-129. In addition marking specfied in paragraph 5.2.2.1 shall apply.

6. NOTES

6.1 The plastic blood bag referred to in 3.4.7 is Federal stock number 6515-664-2734, BLOOD COLLECTING-DISPENSING BAG AND DONOR SET, with Anticoagulant, 450 cc, 4s.

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6.2 Availability of color standards. The color standards of the color referenced in 3.6 are available from:

Commanding General
Aberdeen Proving Ground, Maryland
Attention: Coating and Chemical Laboratory

6.3 Blood bottles and bags for use in the test procedure specified in 4.4.1, shall be supplied by the Government.

6.4 The method of coating the wire racks is commonly known as the Fluidized Bed Process of the Polymer Corporation of America. White cellulose acetate butyrate is available from Eastman Chemical Products Inc., Catalog No. 522G-S.

6.5 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When first article is not required (see 3.2).
(c) Where first article shall be sent (see 3.2).
(d) Level of packing (see 5.2).

6.6 The magnetic vinyl gasket is available from Jarrow Products Inc., 2000 North Southport Avenue, Chicago, Illinois 60614 as Catalog No. P-1943.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review Activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Category: C

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